RMO– 2016(4) Delhi Region

Max Mark – 102 Time – 3 Hrs

Instructions:

- Calculators (in any form) and protractors are not allowed.
- Ruler and compass allowed.
- Answer all the questions.
- All questions carry equal marks.
- Answer to each questions starts on a new page. Clearly indicate the question number.

1. Given are two circle w_1, w_2 which intersects at points X, Y. Let P be an arbitrary point on w_1. Suppose that the lines PX, PY meet w_2 again at points A, B respectively. Prove that the circumcircles of all triangles PAB have the same radius.
2. Consider a sequence $(a_k)_{k\geq 1}$ of natural numbers defined as follows: $a_1 = a,\ a_2 = b$ with $a, b > 1$ and $\gcd(a,b) = 1$ and for all $k > 0$, $a_{k+2} = a_{k+1} + a_k$. Prove that for all natural numbers n and k; $\gcd(a_n, a_{n+k}) < \dfrac{a_k}{2}$.
3. Two circles C_1, C_2 intersects each other at points A and B. Their external common tangent (closer to B) touches C_1 at P and at Q. Let C be the reflection of B in line PQ. Prove that $\angle CAP = \angle BAQ$.
4. Let a, b, c be positive real numbers such that $a + b + c = 3$. Determine with certainty, the largest possible value of the expression $\dfrac{a}{a^3 + b^2 + c} + \dfrac{b}{b^3 + c^2 + a} + \dfrac{c}{c^3 + a^2 + b}$.
5.
   (a) A 7-tuple $(a_1, a_2, a_3, a_4, b_1, b_2, b_3)$ of pairwise distinct positive integers with no common factor is called a shy tuple if $a_1^2 + a_2^2 + a_3^2 + a_4^2 = b_1^2 + b_2^2 + b_3^2$ and for all $1 \leq i, j \leq 4$ and $1 \leq k \leq 3$, $a_i^2 + a_j^2 \neq +b_k^2$. Prove that there exists infinitely many shy tuples.

   (b) Show that 2016 can be written as a sum of square of four distinct natural numbers.
6. A deck of 52 cards is given. There are four suites each having card numbers 1, 2, 3, … ,13. The audience chooses some five cards with distinct numbers written on them. The assistant of the magician comes by, looks at the five cards and turns exactly one of them face down and arranges all five cards in some orders. Then the magician enters and with an arrangement made beforehand with the assistant, he has to determine the face down card (both suit and number). Explain how the trick can be completed.

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